



PS0-ML as you need it

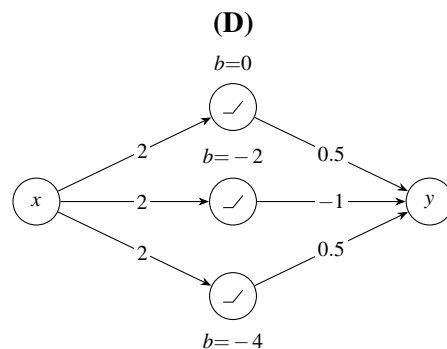
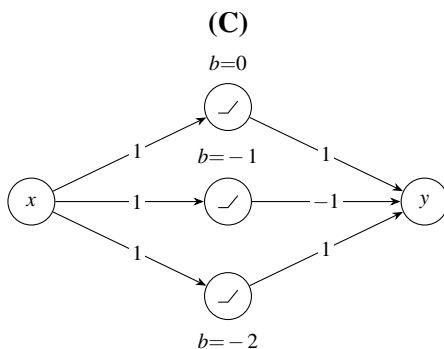
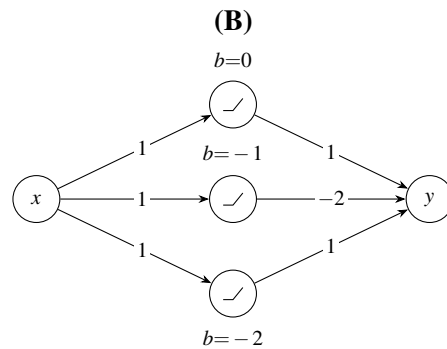
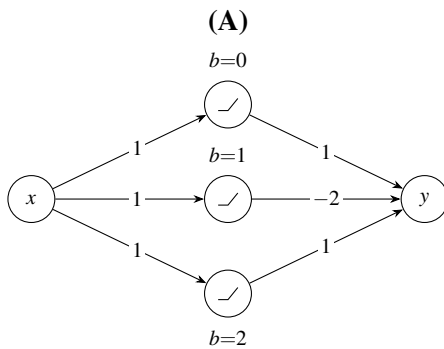
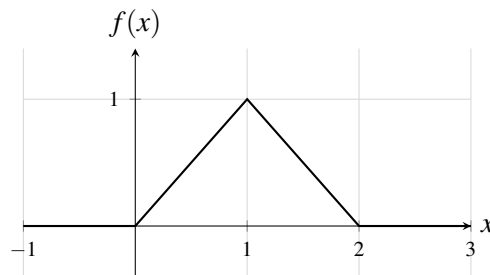
Machine Learning Foundations: Problem Set 0

CSESA

1 Neural Networks

Problem 1.

The plot below shows a target function $f(x)$. Each of the four networks that follow has one input x , three hidden ReLU units, and one linear output unit. A hidden unit computes $h_i = \text{ReLU}(w_i x + b_i)$, where $\text{ReLU}(z) = \max(0, z)$, and the output is $y = v_1 h_1 + v_2 h_2 + v_3 h_3$. The numbers on the edges are the weights, and b is the bias of the hidden unit.



Select all networks whose output equals $f(x)$ for every real x . Justify each choice (correct or incorrect) in one line.

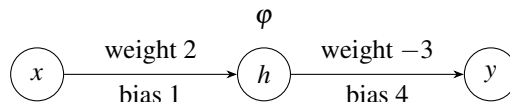
Hint: A ReLU network draws a bent line. The bends occur exactly where some hidden unit's input $w_i x + b_i$ crosses zero.

Problem 2.

A tiny network has one input x , one hidden unit h , and one output y :

$$h = \varphi(2x + 1), \quad y = -3h + 4,$$

where the output unit is linear and φ is the activation of the hidden unit.

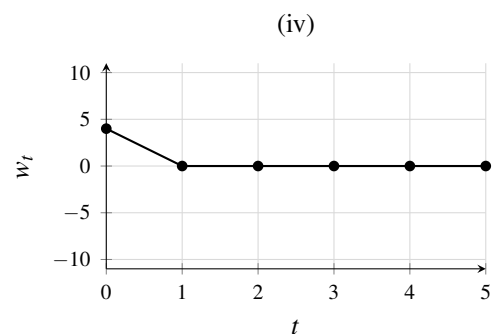
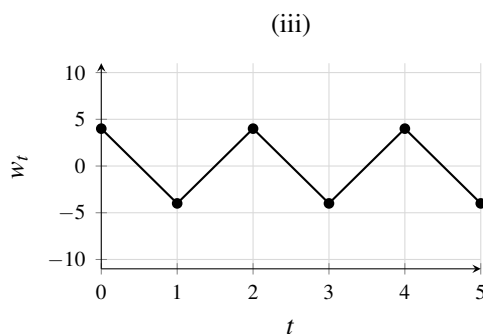
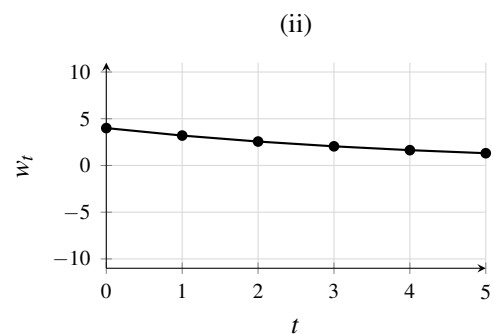
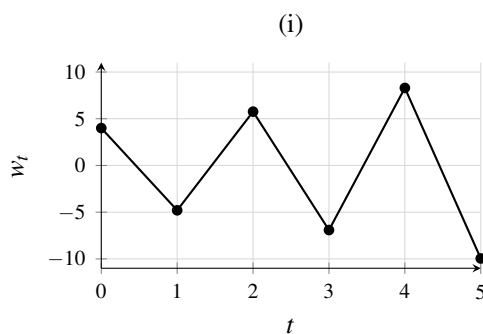


- Let φ be the identity, $\varphi(z) = z$. Show that the whole network collapses to a single line $y = ax + b$, and find a and b .
- Now let $\varphi = \text{ReLU}$. Compute y for $x = -1, 0, 1$, and compare with the line from part (a).
- For which values of x do the ReLU network and the line from part (a) give identical outputs? Describe the ReLU network's output for all other x .
- In one sentence, explain why a deep network needs a nonlinear activation between its layers.

2 Gradient Descent

Problem 3.

Gradient descent is used to minimize $L(w) = w^2$ starting from $w_0 = 4$. The update rule is $w_{t+1} = w_t - \eta L'(w_t)$, where η is the learning rate. The four plots below show w_t for $t = 0, \dots, 5$ for four different learning rates $\eta \in \{0.1, 0.5, 1.0, 1.1\}$, in scrambled order.



- (a) Show that the update rule becomes $w_{t+1} = (1 - 2\eta)w_t$.
- (b) Match each plot to its learning rate, using the rule from part (a).
- (c) Which of the four learning rates make w_t converge to the minimum $w = 0$? For what range of η does this happen in general?
- (d) Plot (iii) neither converges nor blows up. Explain what the algorithm is doing.

Problem 4.

A line $\hat{y} = wx + b$ is fitted to two data points $(x, y) = (1, 2)$ and $(3, 4)$ using the mean squared error

$$L(w, b) = \frac{1}{2} \sum_{i=1}^2 (wx_i + b - y_i)^2.$$

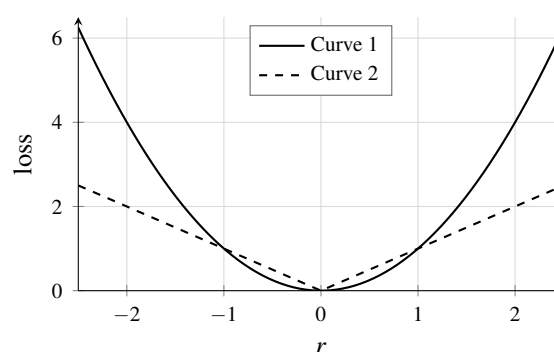
It can be shown that $\frac{\partial L}{\partial w} = \sum_i (wx_i + b - y_i)x_i$ and $\frac{\partial L}{\partial b} = \sum_i (wx_i + b - y_i)$. Start from $w = 0$, $b = 0$ and use $\eta = 0.05$.

- (a) Compute the loss $L(0, 0)$.
- (b) Compute $\partial L / \partial w$ and $\partial L / \partial b$ at $(0, 0)$.
- (c) Perform one gradient descent step to obtain the new (w, b) .
- (d) Compute the new loss and verify that it decreased. What would have happened to the loss if you had added the gradient instead of subtracting it?

3 Loss Functions

Problem 5.

Two loss functions are drawn below as a function of the residual $r = \hat{y} - y$, the prediction minus the true value. One of them is the squared error r^2 and the other is the absolute error $|r|$.



- (a) Identify Curve 1 (solid) and Curve 2 (dashed) as squared error or absolute error.
- (b) A model predicts a constant value c for the data $\{1, 2, 3, 4, 100\}$. Find the value of c that minimizes the mean squared error, and the value that minimizes the mean absolute error. (*Hint: think of the mean and the median.*)
- (c) Which loss is more sensitive to the outlier 100? Use the shapes of the two curves to explain why.

Problem 6.

Three patients all truly have a disease (label $y = 1$). A classifier outputs p , its predicted probability that $y = 1$. It predicts “disease” when $p \geq 0.5$. For a true label $y = 1$, the cross-entropy loss of one example is $-\ln p$. Two models are compared:

	Patient 1	Patient 2	Patient 3
Model A, p	0.51	0.51	0.51
Model B, p	0.99	0.99	0.49

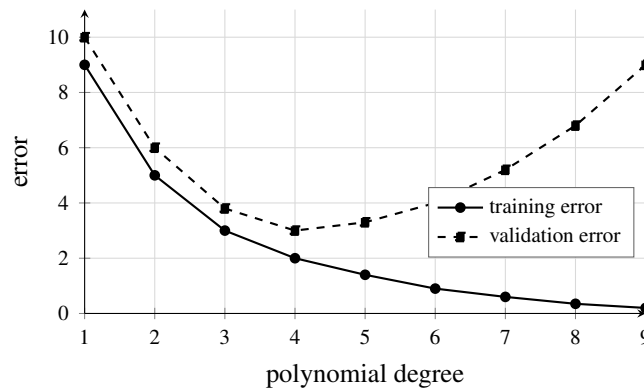
p	0.99	0.51	0.49
$-\ln p$	0.010	0.673	0.713

- Compute the accuracy of each model.
- Compute the mean cross-entropy loss of each model, using the table of $-\ln p$ values.
- Which model has the higher accuracy, and which has the lower loss? Explain in two or three sentences why accuracy and cross-entropy can disagree.
- A third model outputs $p = 0$ for a patient who truly has the disease. What is the cross-entropy loss, and what does that say about overconfident wrong predictions?

4 Basic Ideas of Machine Learning

Problem 7.

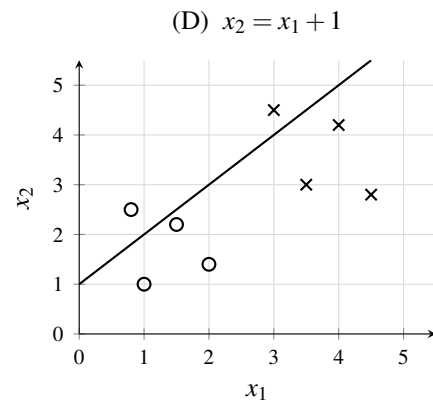
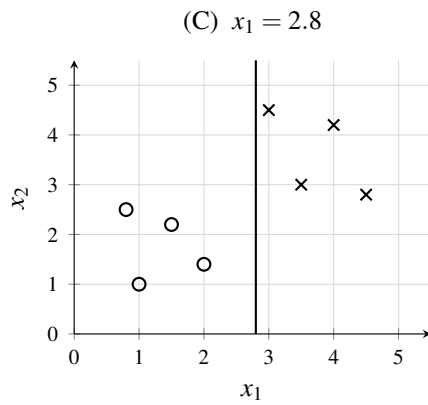
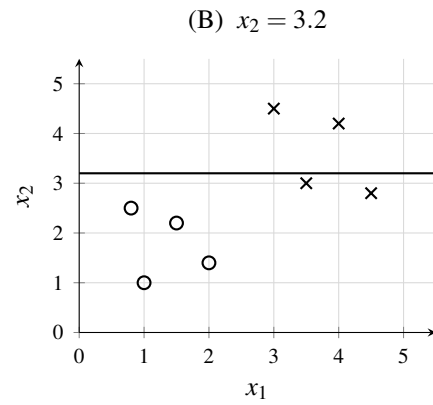
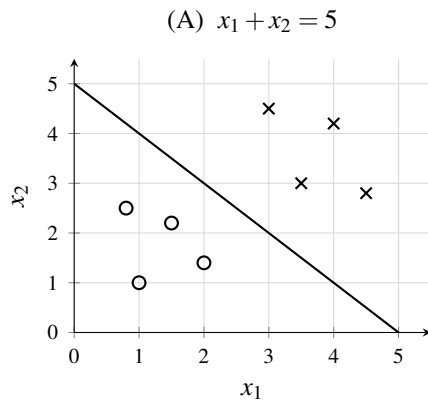
A student fits polynomials of degrees 1 to 9 to the same dataset. The plot shows the error on the training data and the error on a separate validation set, both measured after fitting.



- Which degree would you choose, and why did you not choose the degree with the lowest training error?
- Degree 1 and degree 9 both have high validation error. Name the problem each one suffers from (underfitting or overfitting) and explain in one sentence.
- Why should the final performance of the chosen model be reported on a third, untouched test set instead of the validation set?

Problem 8.

Each point below is a patient with two measurements, x_1 and x_2 . A circle $\circ$ means “healthy” and a cross $\times$ means “ill”. A linear classifier draws a straight line and predicts “ill” on one side of it. Four candidate lines are shown on the same training data.



- (a) Select all lines that classify every training point correctly. Name one point that each wrong line misclassifies.
- (b) Is this task classification or regression? Is it supervised or unsupervised? What are the *features* and what is the *label*?
- (c) The classifier is $w_1x_1 + w_2x_2 + b = 0$. Which of the following are *parameters* (learned from data) and which are *hyperparameters* (set by the programmer): w_1 , b , the learning rate η , the number of training epochs?